

Final Term Paper Submission

The Spatio-Temporal Geopolitics of Renewable Energy Loans and the Struggle for Influence: China and Africa

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Introduction – Increasing Carbon Emissions

The world is experiencing climate change and carbon emissions are being identified as the major cause for this. Scientists have identified carbon emissions to be of either of a natural or anthropogenic origin; while there have been fluctuations in natural carbon emissions that have led to unfavorable circumstances over millions of years, recent human activities have resulted in a huge spike in carbon emissions and temperatures globally. Figure 1 and Figure 2, highlight how emissions have been rising rapidly in India and China since the 1980s. It has now been globally recognized that climate mitigation action needs to be taken; however, large lump sum investments and major technological breakthroughs are required to achieve the same. At present, developing countries aim at industrializing rapidly to achieve higher growth rates, while on the other hand, developed countries aim at strengthening their supply chains and moving industries out to developing countries passing on the carbon burden to them. This paper tries to discover the intersection between developed and developing countries, their diplomatic agreements and shift from fossil fuels to clean energy; in particular for China, which has not only emerged as the highest lender for loans in renewable energy projects, but also the only developing country to have the capacity to give loans for renewable energy projects to any country.

Renewable energy is a technological solution to reduce carbon emissions. Not only will a clean energy transition help mitigate climate change, it will also help build resilience against volatile energy prices and improve energy supply to the rural areas. The creation of renewable energy microgrids in rural areas will ensure supply of electricity to the villages at cheap rates; however, any renewable energy project comes with very high investment. Developing countries, hence, rely on foreign aid, either in the form of grants or loans to finance renewable energy projects. The primary donors are either international organizations or other developed nations, with the exception of China.



Power Diplomacy – Developed Nations on top of Developing Nations

The climate change crisis is a global issue which requires cooperative action; however, developed countries have been accused of shifting their responsibilities to the developing nations -- greater power allows them to drive agreements and craft exemptions. The United Nations Framework Convention on Climate Change (UNFCCC)'s conception of common but differentiated responsibilities and respective capabilities (CBDRRC) is a principle which acknowledges differing responsibilities to be fulfilled in accordance with the national capacities; it rather back fired giving the developed nations a "free to pollute" card by means of investing in renewable energy projects in developing nations and carrying out high emission production processes within.

The developed nations have also been accused of "Carbon Leaking" -- A practice wherein developed nations have set up factories in developing nations to manipulate the process of carbon counting; carbon is counted on a production level as opposed to where the goods/services are consumed; thereby, shifting manufacturing activities to countries with lower norms for environment protection -- creating a detrimental impact on the environment. A possible reasoning for the opposition to a clean energy transition is the fact that the developed countries have already invested heavily in traditional energy sources, such as coal, oil, and natural gas, and have established infrastructure for these sources; the transition to renewable energy sources would require significant investment and restructuring of existing infrastructure -- This may be less attractive than continuing to use traditional energy.



Flip side – Developing Nations Prioritize a Clean Energy Transition

On the flip side, renewable energy is becoming increasingly attractive for developing countries. Developing countries, unlike developed countries, lack the existing infrastructure to meet the energy needs of the population using fossil fuels. Rural areas, in particular, are affected the most. Investment in renewable energy projects will provide options for rural areas that lack electricity transmission and distribution infrastructures and thereby lifting individuals out of energy poverty. One can also draw possible conclusions from a Jared Diamond point of view; developing countries are endowed with an abundance of natural resources -- making it easier to harness the power of renewable energy. Additionally, renewable energy projects generate employment, establish goodwill, enhance energy security, and prevent balance of payments burdens since there exists little to no volatility in energy prices from renewable energy. The future costs to supply renewable energy is also expected to fall. Finally, it's seamless alignment with the process of industrialization building upon the forward and backward linkages makes it attractive to the developing countries.

Data and Methodology-- The answer lies Within

This paper aims at looking at whether there exists a pattern among the loans given for renewable energy projects by China to the world; in particular looking at whether China gives loans to countries classified as “lower income countries” by the world bank. This paper also aims at comparing India, China and Africa to come up with noteworthy observations regarding the inflow and outflow of loans, grants and equity in these regions. The Spatiotemporal framework has been followed for this paper, looking at the differences between developed countries and developing countries over the years. This paper uses the investment in renewable energy projects data, which is a publicly available dataset on IRENA (International Renewable Energy Association) and conducts data wrangling operations on it to highlight trends. Additionally, Income Classification Dataset and GDP (PPP) data have been used in order to strengthen the arguments made in the paper.

Debts or Grants -- The Current state of Renewable Energy Projects

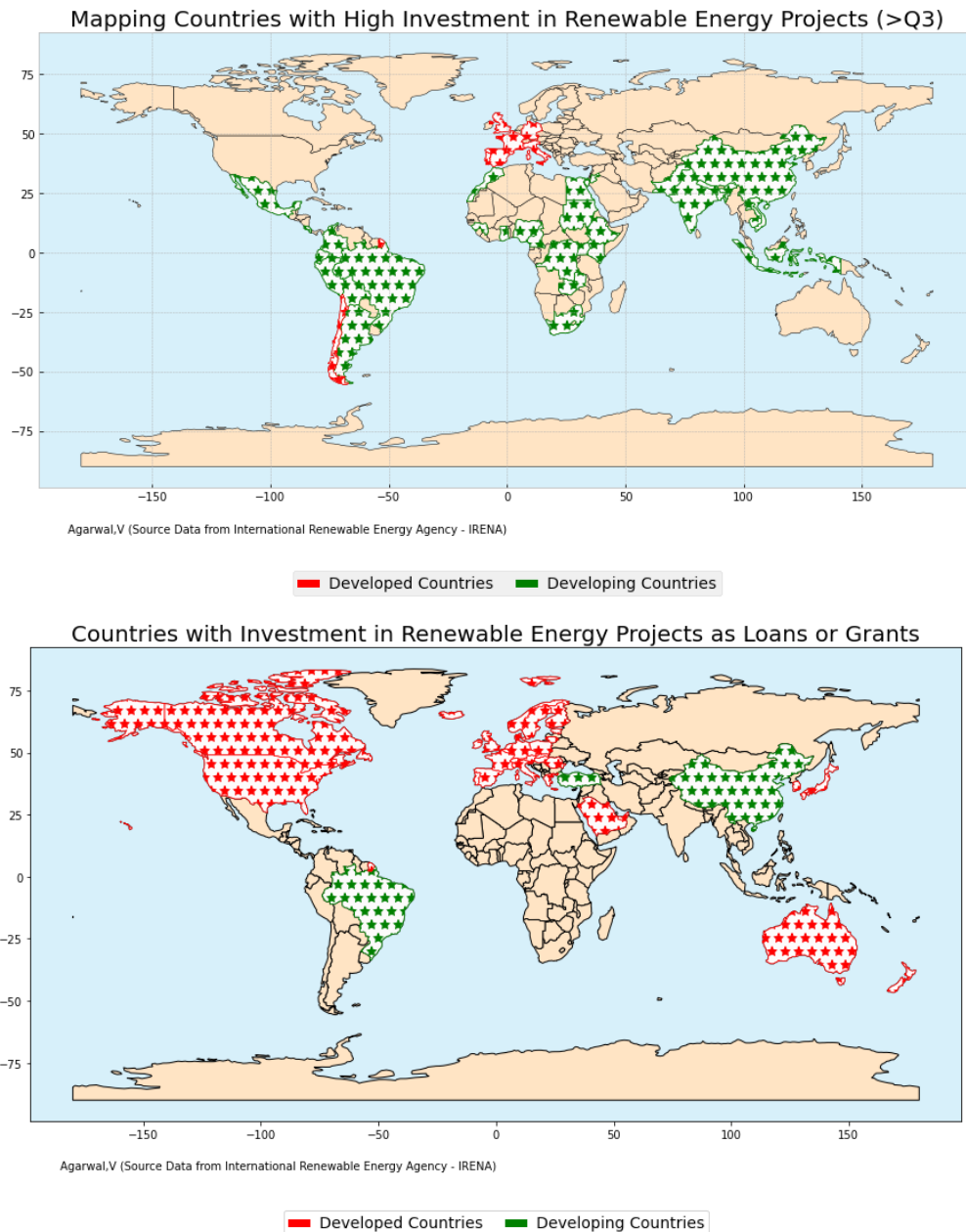


Figure 1: Contrasting difference between countries where Renewable Energy Projects are being established (Developing nations) and where the funding comes from (Developed nations with the exception of China)

Developing countries have received massive (massive – in this case, is categorized by investments in projects falling above the 3rd quartile) investments to improve the state of the renewable energy projects financed either through loans, grants or equity investments. Nearly all countries, with the expectation of western european countries, don't belong to the world bank income classification of “high income countries”. China is the only expectation¹ to the norm of developed countries funding projects in developing nations. This indicates that funds are flowing from developed and prosperous countries to developing countries. The next step is to look at how developing countries are funding these projects and to whom.

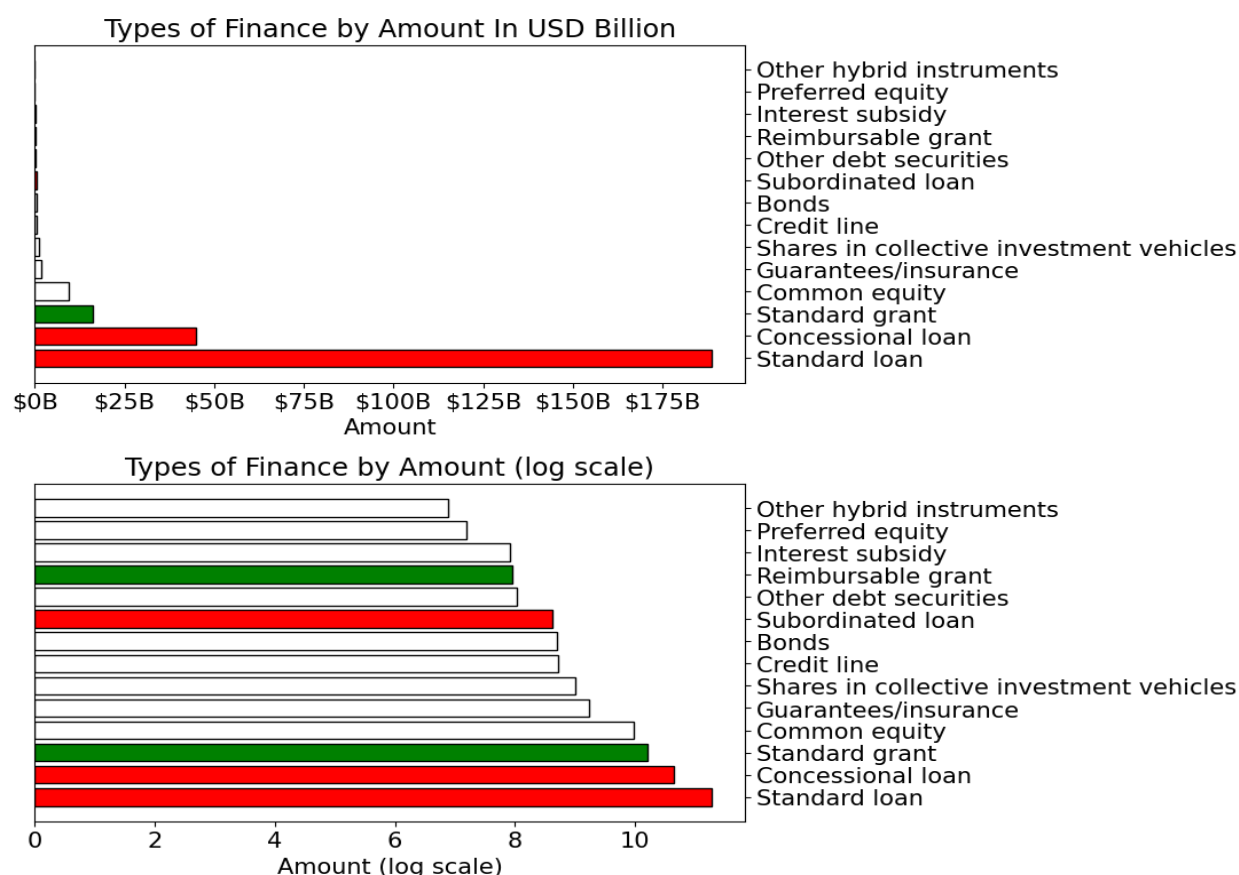


Figure 2: Renewable Energy Projects are financed primarily through loans. (Both graphs show the same values; the second facet is log transformed)

Most projects are financed through loans given by developed countries to developing countries. The only expectation, China, turns out to be the largest donor to support renewable energy projects in developing countries; China also lends only through loans. Table 1, shows the split of

¹ The Turkish and Brazilian governments both invest in funds set up to aid the clean energy transition in their respective countries and do not lend to any other country; Making China the only “Non-High Income country to invest in renewable energy projects.”

amounts given as loans, grants and equities by the top ten countries or institutions. The values in parentheses correspond to the amount lent in million US Dollars.

Top Donors (Countries/Organisations)	Top Donors by Loan	Top Donors by Equity	Top Donors by Grants
China (54594.44)	China (54594.44)	United Kingdom (3601.24)	Germany (2644.67)
Brazil (43369.8)	Brazil (43367.62)	International Finance Corporation (2059.49)	EU Institutions (2431.58)
EU Institutions (27289.73)	EU Institutions (24236.87)	Norway (633.21)	International Development Association (1555.88)
Germany (15626)	Germany (12642.18)	Islamic Development Bank (510.33)	Japan (892.13)
International Finance Corporation (14827.66)	International Finance Corporation (12527.29)	EU Institutions (502.54)	United States (876.89)
Japan (12621.66)	International Bank for Reconstruction and Development (11734.62)	Netherlands (489.02)	Inter-American Development Bank (769.81)
International Bank for Reconstruction and Development (11734.62)	Japan (11538.24)	Asian Development Bank (464.99)	Asian Development Bank (695.73)
United States (10515.72)	United States (7968.35)	Inter-American Development Bank (452.84)	Netherlands (689.75)
Asian Development Bank (8727.04)	Asian Development Bank (7566.32)	European Bank for Reconstruction and Development (448.77)	Norway (630.85)
International Development Association (7396.77)	International Development Association (5840.89)	Green Climate Fund (424.71)	United Kingdom (613.73)
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Table 1: China has lent money to other countries through only loans

The Borrower and the Lender – Comparing India and China

Top Donors to China	Amount in USD Millions
Asian Development Bank	1089.80
International Bank for Reconstruction and Development	1062.5
International Finance Corporation	725.59
EU Institutions	655.23
Japan	488.62

Top Donors to India	Amount in USD Million
Germany	2468.63
Asian Development Bank	2371.82
International Finance Corporation	1912.45
International Bank for Reconstruction and Development	1897.41
United States	1480.32

Table 2: Comparing Indian and Chinese Lenders.

Both India and China, over the recent years, have taken renewable energy seriously. There has been a significant increase in the number of operational renewable energy plants in both these nations. While India on one hand has taken more significant levels of loans to finance renewable energy projects in India, it has still not lent loans to any other country to help them finance their renewable energy needs; China, has aggressively given loans to other countries, while expanding

their domestic renewable energy capacities as well. This gives India the status of a borrower and China a status of a lender, as it relies mainly on large foreign investment in the form of loans to finance renewable energy projects. Till date, there has been no interaction between India and China on renewable energy projects; however, the Chinese development bank has lend a loan to India twice, one in 2010 and again in 2013 for Coal related projects.

Caught RED Handed -- Africa, China and the World a Visual Tour

Top Debtors - China	Top Debtors - EU	Top Debtors - Germany	Top Debtors - US	Top Debtors - Japan
Nigeria (7025.42)	Germany (3167.11)	India (2468.68)	India (1480.32)	Indonesia (1813.93)
Lao People's Democratic Republic (6560.75)	United Kingdom of Great Britain and Northern Ireland (3078.92)	Morocco (2221.96)	Chile (1377.42)	United Kingdom of Great Britain and Northern Ireland (1576.36)
Ethiopia (5842.66)	France (1985.83)	Brazil (925.49)	Kenya (989.74)	India (1399.01)
Pakistan (4490.68)	Spain (1908.41)	Unspecified, developing countries (910.35)	South Africa (753.16)	Kenya (961.38)
Ecuador (4000.42)	Portugal (1601.66)	Indonesia (849.07)	Peru (606.25)	Egypt (883.35)

Table 3: China has lent a significant amount of money to economically backward or countries belonging to the African region.

Top Debtors to Low Income Countries	Amount in USD Million
China	14638.07
International Development Association	1548.47
France	652.24
International Finance Corporation	607.32
Netherlands	301.32

Table 4: China has lent the highest amount, by a very large margin to low income countries.

Looking at Top Donors in the African Region by Loan, Equity and Grants.

Top Donors	Amount in USD Million	Ratio to Chinese Investment
China	28189.42	1
Germany	3933.34	7.17
International Finance Corporation	2858.43	9.86
France	2737.62	10.3
International Development Association	2674.26	10.54

Top Doners (Equity)	Amount in USD Million	Ratio to Chinese Investment
1 Norway	452.43	62.31
2 EU Institutions	355.42	79.31
3 International Finance Corporation	352.27	80.02
4 Germany	160.74	175.38
5 Green Climate Fund	143.81	196.01

Top Donors (Grants)	Amount in USD Million	Ratio to Chinese Investment
EU Institutions	1325.82	21.26
International Development Association	1196.88	23.55
Germany	1036.78	27.19
United States	600.99	46.9
United Arab Emirates	321.37	87.72

Table 5: Looking at Loans, Equity and Grants in Africa, China lent far more for renewable energy projects to Africa than any other country

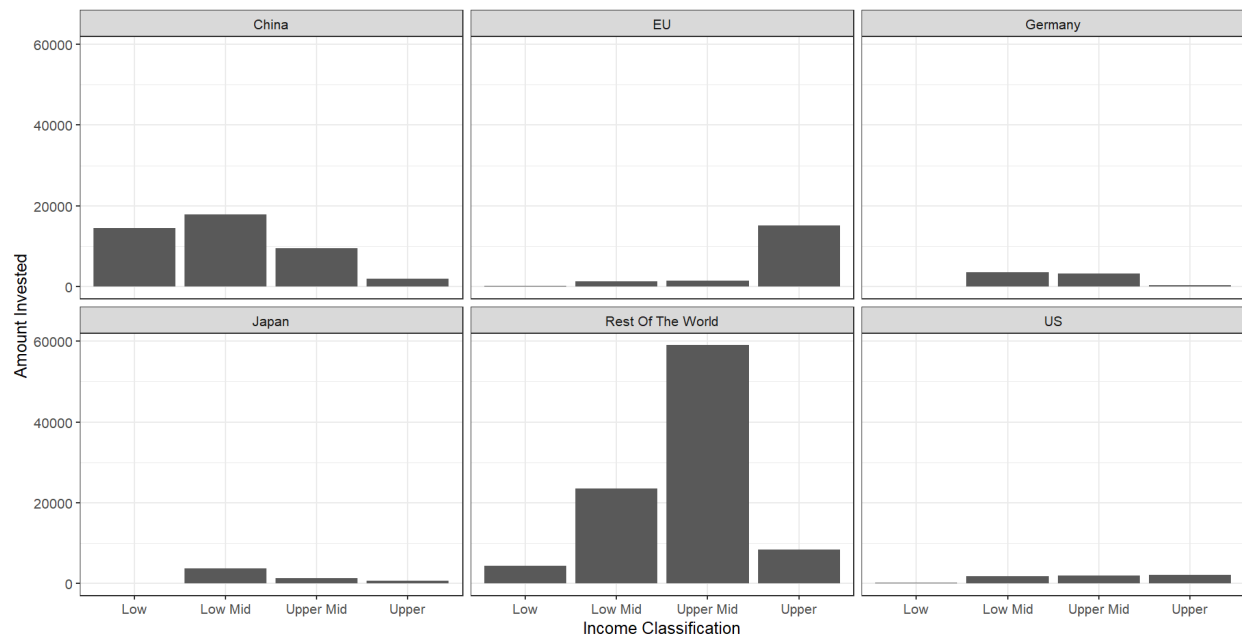


Figure 3: Grouping the investments in renewable energy projects by the world bank classification we can see that China has clearly lent a significant amount of loans, far greater than any other country to low income countries.

A Twist? Or Twisted? -- Discussion

The IRENA dataset based on investment in renewable energy projects showed plenty of interesting observations – While plenty of renewable energy projects are emerging in developing countries, these projects are most often financed by developed nations through loans. It is usually the developed nations which have provided loans, with an exception of China. Not only is China the only developing country to give loans to other countries for renewable energy projects, it is also the country with the largest amount of loans given. It is clear that China has been giving loans to economically poor countries; however, more work needs to be done in order to link this trend with debt trap diplomacy – a practice China has been accused of recently. In my opinion, I do believe that the motive of this move would be to gain greater influence; however, this requires an in depth study of the BRI (Belt and Road Initiative) which would help base stronger arguments, which is a future scope of work.

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Datasets used

- Public Investment in Renewable Energy Projects July14 2022, IRENA
- World Bank Income Classification and GDP (PPP) Data, World Bank.

